B - Mechanical Design MS4092

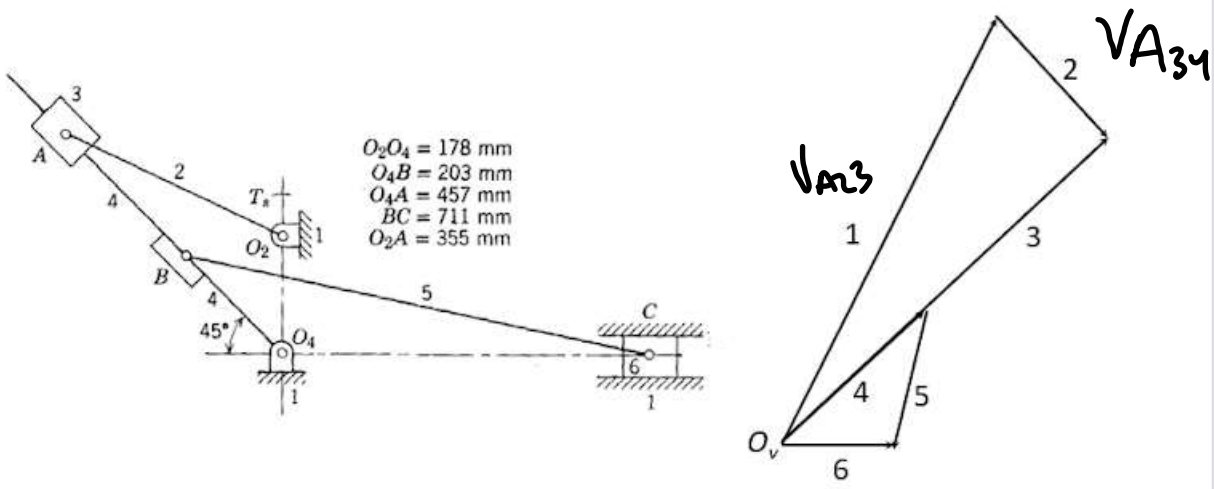
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Number of problems: 8
Start at: 14:30
Finish at: 14:54
Duration: 24 Minutes

When a system vibrates in the rough surface, the damping is...

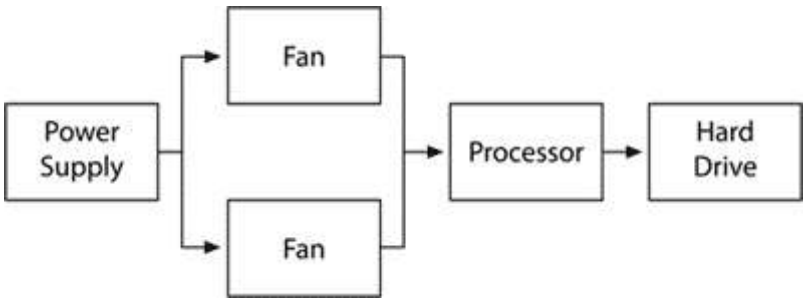
- ○ Solid
- ○ Hysteretic
- ⊗ Coulomb
- ○ Viscous

Look at the following diagram of the mechanism and velocity polygon! If link 2 is rotating in clockwise direction, which below statement is true?



- ○ Vector number 2 is V_A4A3
- ○ Vector number 1 is V_A4
- ⊗ Vector number 5 is V_CB
- ○ Vector number 3 is V_B

A Reliability Block Diagram (RBD) of a simplified computer system with a redundant fan configuration is shown below. If the reliability of each component is 0.9, what is the system reliability?



- 0.49
- 0.41
- 0.81
- 0.72

$R_p = 1-[(1-0.9)(1-0.9)] = 0.99$

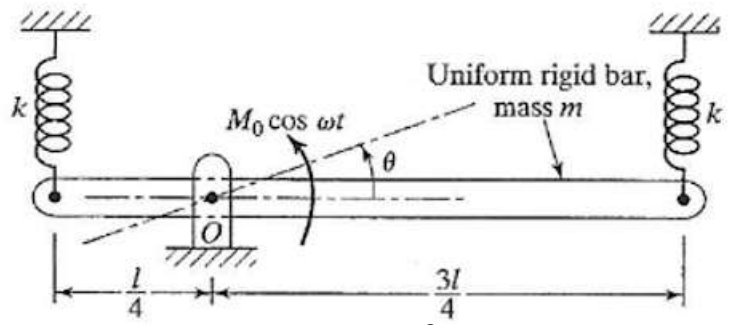
$R_{Tot} = 0.9 \times 0.99 \times 0.9 \times 0.9 = 0.72$

The Frequency Response Function (FRF) represents dynamic characteristic of a system for the following conditions, except:

- Transient state
- The control system is an closed-loop system
- The system is stable
- Sinusoidal inputs

Please answer the following problem.

The equation of motion of the system shown in the figure below for rotational motion about the hinge O is $A\ddot{\theta} + B\theta = C\cos(Dt)$. It is known that $k = 4000$ N/m, $l = 1$ m, $m = 10$ kg, $M_0 = 100$ Nm, $\omega = 100$ rad/s, and $J_o = \frac{7}{48}ml^2$. The value of A, B, C, and D of motion equation are…



- A = 7, B = 1200, C = 480, D = 100
- A = 7, B = 15000, C = 480, D = 100
- A = 7, B = 12000, C = 480, D = 100
- A = 7, B = 1500, C = 480, D = 100

$J_o \ddot{\theta} + \frac{10kl^2}{16}\theta = M_0 \cos\omega t$

$\frac{7}{48} \cdot 10 \cdot 1^2 \ddot{\theta} + \frac{10 \cdot 4000 \cdot 1^2}{16}\theta = 100\cos 100$

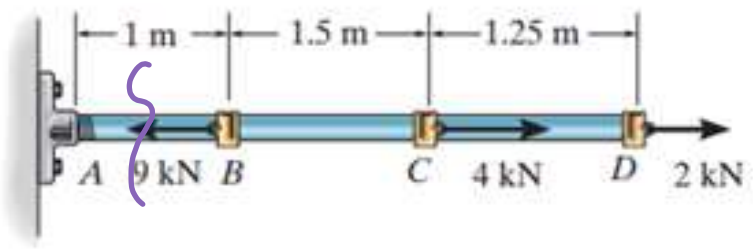
$A = 7$

$B = 12000$

$C = 480$

$D = 100$

The A-36 steel rod (E = 200 GPa) is subjected to the loading shown. If the cross-sectional area of the rod is 50 mm^2 determine the maximum shear stress of the segment AB. Neglect the size of the couplings at B, C, and D.



$SF = 6 - 9 = -3\ kN$

- [x] 60 MPa
- [ ] 40 MPa
- [ ] 120 MPa
- [ ] 30 MPa

$$\sigma = \frac{3 \times 10^3\ N}{50 \times 10^{-6}\ m^2}$$

$$\sigma = \frac{3 \times 10^9}{50} \cdot \frac{1}{10^6}\ MPa = 60\ MPa$$

Please answer the following problem

A system has transfer function of

$$G(s) = \frac{20}{s(s+1)(s+20)}$$

Incorrect statement regarding the system is:

- [ ] If a unit step input is applied to the system of G(s), then the final value of the response is 1 unit.
- [x] Static gain of G(s) is 1
- [ ] The step response of G(s) is slower than the step response of $G_2(s) = \frac{20}{s(s+20)}$
- [ ] The dynamic characteristic of G(s) is similar with dynamic characteristic of $G_1(s) = \frac{1}{s(s+1)}$

A refrigerator of weight 800 N is on the rough floor with static friction coefficient μs = 0.25. A man pushes the refrigerator in the horizontal direction as shown in the figure below. His weight is 750 N and the static friction coefficient between his shoes and floor is μs = 0.6. The correct statement below about the refrigerator is...

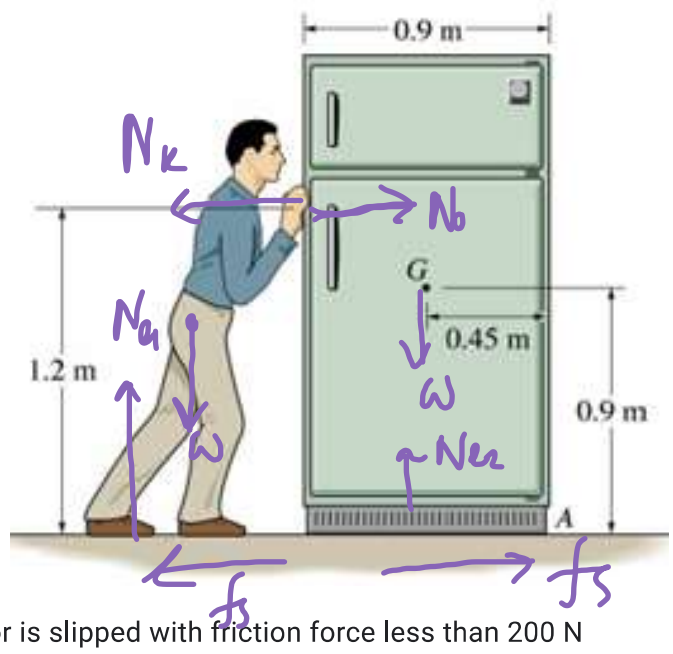
orang: $N_o = N_k$

$f_{so} = \mu_s . N_{R_1} = \mu_s . w$

$= 0,6 \times 750 = 450\ N$

$N_k . 0,6 = f_s . 0,6$

$N_k = 450\ N$



- The refrigerator is slipped with friction force less than 200 N
- The refrigerator is tipped with friction force 200 N
- The refrigerator is slipped with friction force 200 N
- The refrigerator is tipped with friction force less than 200 N

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kuliah: $N_{R_2} = w = 800\ N$ $\mu_s = 0,25$

$f_s = 0,25 \times 800 = 200\ N$

$N_o . 0,3 = f_s . 0,9$

$f_s = \frac{450}{3} = 150\ N$